

15 Share sustenance: Bridging food surplus and insecurity for a sustainable future

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Abstract: Food sharing and tracking is an innovative approach to tackling food waste and ensuring efficient resource allocation. In a world where sustainability is paramount, food surplus remains a critical issue. Our Android application, named “Share Sustenance,” is designed to connect individuals and businesses to securely share excess food, thereby reducing waste and addressing food insecurity. Human dignity and well-being require sufficient nutritious food, but worldwide food insecurity continues to exist. The solution demands both technological advancements and active participation of local communities. In the world of technical advancement in India, we need to build the technical solution which are less costly and most efficient. Our application enhances food donation by providing user benefits that drive active food donations while enabling volunteer contribution tracking through real-time monitoring which builds trust through transparent delivery confirmation. This platform promotes social connections among community members while encouraging food sharing to transform distribution methods for better environmental impact and social benefits. The solution provides an efficient technology-based system which bridges food surplus with scarcity to build a sustainable food-secure future. This application also features a Real-time tracking from user, admin and delivery person end to promote transparency.

Keywords: Food sharing, food waste reduction, sustainability, community engagement, food tracking, volunteer donations

1. Introduction

While a hunger-related death list ranks at 95, the United Nations reports that India wastes about 7.5 tons of food every day. Giving weddings, ropes, and big show days will have extra food that, most of the time, stays unused, fostering food insecurity and environmental harm. Food wastage depletes the edible food and creates potential carbon emissions; hence, it stands as one of the global concerns. With the advancement of technology, AI, and digital platforms, many food-sharing apps have been created and launched to interlink individuals, businesses, and nonprofits for food redistribution. But less than 20% of people are actually donating food using such platforms. Lack of awareness and incentives possibly stifles the potential reach of existing programs. People are still not trusting in such applications, and hence it is important to not only build the solution but also promote the same. This paper is to develop an app for Food Sharing and Tracking; it facilitates food donations seamlessly and incentivizes users through a “Food Coins” reward policy. These coins could be redeemed for vouchers, thereby encouraging more people and companies to engage in food sharing. Also, there

will be a tracking function available on the app, so the users may track the extent of food redistribution, reduction of food-related wastes, and environment impacts. Technological plus community incentives are applied so that this activity would stimulate engagement, limit food wastage, and promote sustainability at a large scale.

2. Literature Survey

With technology and digital platforms evolving at a rapid pace, the processes of food donation and waste management have simultaneously faced revolutionary changes. This research presents an investigation into several food donation systems across diverse approaches including peer-to-peer sharing, large-scale redistribution, and community-driven initiatives. From the research, it is evident that technology-driven donation systems might boost food sharing efficiency and strengthen food security.

[1] M. Banerjee and S. Chatterjee, SeVa: A Food Donation App for Smart Living, IEEE, 2023.

This research paper introduces Zero Hunger: Smart Food Donation System using IoT, which provides three

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modules – donor, admin, and volunteer – to facilitate food donations. Donors register, authenticate via mobile verification, and list food donations, which are then assigned to NGOs for collection.

[2] P. Kumar and M. Reddy, An Android Application Development for Food Donation: A Geographical Location-Based Approach, IEEE, vol. 12, no. 4, 2023. This research paper presents a Food donation application focusing on reducing the wastage of food and daily utensils. The solution consists of 2 modules that is, donor and receiver, both verify using OTP verification.

[3] D. Bose, L. Fernandez, A. Rao, “A Centralized Platform for Bridging the Gap between Donors and Recipients,” IEEE Transactions on Humanitarian Technologies, vol. 15, no. 3, pp. 345–358, 2023.

[4] S. Gupta, R. Verma, and A. Sharma, created an examination solution for the donation of Food, Book and clothes between donors and recipients IEEE, 2023. This research highlights the Share Your Food application, which provides an interactive platform for food donation. Donors log in and input details about food donations, contact information, and location.

[5] E. J. George, A. R. Khan, M.K., M. S. P. T., and RM., FoodShare: A Collaborative Platform for Food Donation and Distribution. This research presents a system for donation of not only food but also daily utensils such as Books, Clothes, and daily required small things.

[6] Akhil Menon and his team developed an Android Application that featured as a system notification which means the user can be aware of the status of the donation. In such a system, the user sends a request, which goes to the NGO, where they can accept donations if required.

[7] As mentioned in [7], Sharma and Gupta proposed an approach that identifies and redistributes surplus food, aiming to enhance efficiency through automated solutions.

[8] A modern technology developed by Menon and his team, emphasizing door-to-door service like from donor to recipient connection via an Android Application.

[9] Brown and Lee created a case study showing how an android application can efficiently bridge the gap between the user and receiver using real-time technology.

[10] Johnson and Patel built an Android application for a social cause, connecting local communities and donors with intuitive features of application.

This research identifies gaps with existing food-sharing systems:

1. Limited Real-Time Tracking and Data Analytics-Most food-sharing apps concentrate on connecting donors with recipients without getting into the real-time food redistribution processes or quantifying food waste reduction.
2. Limited Participation from Individuals and Small Households-The most significant contributors to food donation platforms remain businesses and restaurants,

and there is an underrepresentation of individual households.

3. Proposed System

Currently, food donation online platforms are limited, and people are not properly aware of those systems. Many systems lack transparency and efficiency. Existing systems are mainly focusing on large organizations and corporations for their support and ignoring small households and communities, who also don't get food on a daily basis. This leads to food waste at the community level, where people may wish to donate surplus food but struggle to find an efficient way to do so. This solution aims to provide a simple and easy-to-use food donation and tracking system for Households, communities, and organizations so that everyone can benefit from this solution. This Android application comprises different functions, which include the donation of food, the receiving of food, tracking of food, and maintaining the quality of food. NGOs and food recipients can check available donations and request food. In order to track the donations, redistributions, and the effect on reducing food waste, it incorporates real-time tracking and analytics. Android development technology, which is mostly used across the development process, that is, Java is used in building the solution for the backend process. XML is used for creative layout and is simple to handle and process the pages separately. Firebase is used for real-time database fetching and authentication purposes. Firebase makes the application more secure and reliable. The overall application is built by keeping the research gaps in mind to make the changes in this solution and provide a proper solution for society.

4. Methodology

Several methods have been used in the past to create food donation and tracking systems. These include peer-to-peer food-sharing platforms, large donation networks, and non-perishable food distribution systems. Despite many improvements in this area, current solutions often lack real-time tracking, effective donor engagement, and a well-organized food delivery system. Furthermore, most platforms focus on large businesses or NGOs. This approach limits the involvement of small households and individual donors.

This food donation app aims to make it easier to donate and receive food, clothing, and books. It has three main types of users: Donors, Delivery Personnel, and NGOs. As stated in Figure 15.1, the user interacts with the interface by registering and logging into the system. The model facilitates food donation and distribution through the following steps.

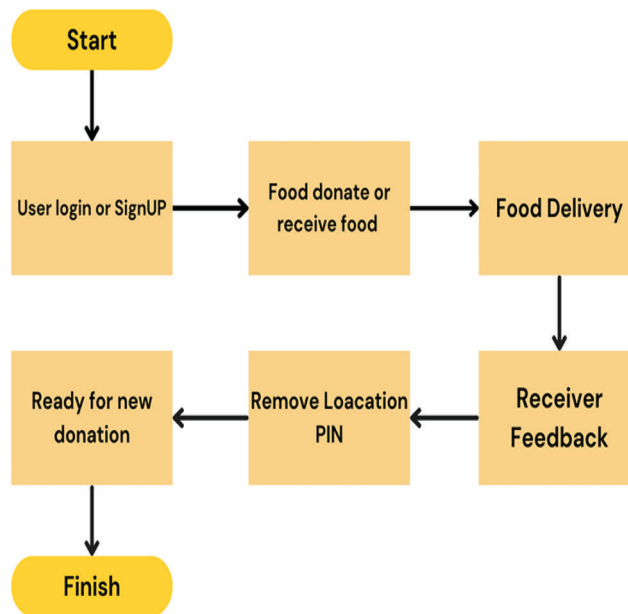


Figure 15.1. Workflow of the model.

Source: Author.

User authentication: After starting the system, the main task is user sign-up or login. Users are required to fill in their login credentials to proceed with the donation process for security purposes.

Food Donation/Receiving: Users have the option to request food as a recipient or donate food. Donors supply information about the type, amount, and location of the food pickup. Depending on availability and proximity, recipients can peruse the available donations.

Food Delivery: After a recipient claims the food, logistics are set up by volunteers, non-governmental organizations, or a self-pickup option. To cut down on waste, the system makes sure food is delivered effectively.

Feedback from Recipients: To ensure openness and quality assurance, users offer feedback following the donation.

Location PIN Removal: Once the food is delivered at the recipient's address, they can remove their location PIN and continue with a different food donation request as needed.

Ready for New Donation: The donor can then proceed with a new donation, completing the cycle.

5. Results

The solution model runs smoothly once started, provides an easily accessible features for user to use. It can easily be downloaded in any device and model as being compatible for all models is also an advantage. The application is designed to facilitated smooth communication between Donor and Receiver. The activity status is easily updated by delivery person for the acknowledgement of users.

Figures 15.2 and 15.3 showcase the user signup and login interface. User authentication is an essential part of

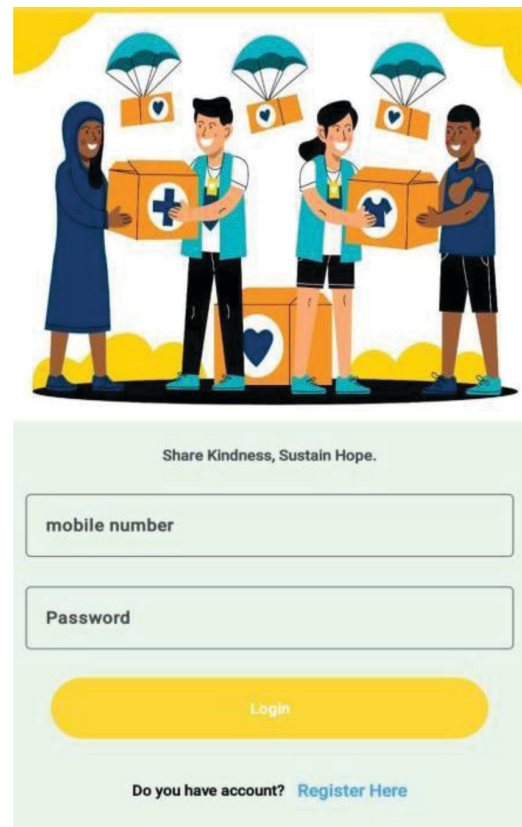


Figure 15.2. User login interface.

Source: Author.

the system. It allows users to register using their login credentials and password for security purposes.

As seen in Figure 15.3, users are required to provide all necessary information such as name, email address, residence address, mobile number, and password. Use of Firebase authentication ensures that only legitimate users can access the login dashboard. The dashboard, as shown in Figure 15.4, has several functions, such as donating food, receiving food, accessing the food map, viewing history, managing credentials, and changing settings. The interface is designed to be user-friendly, making navigation easy for everyone.

Figure 15.4 shows the home dashboard of the donation app. It displays important statistics, including the number of registered NGOs, delivery persons, and total donations completed. It also provides quick donation options for clothes, books, and food, making it easy for users to contribute right away.

Users can track their past donations and received food through the history feature (Figure 15.5). The system stores details of previous transactions, including food type, quantity, and timestamps, allowing users to keep track of their contributions. Additionally, users can manage their

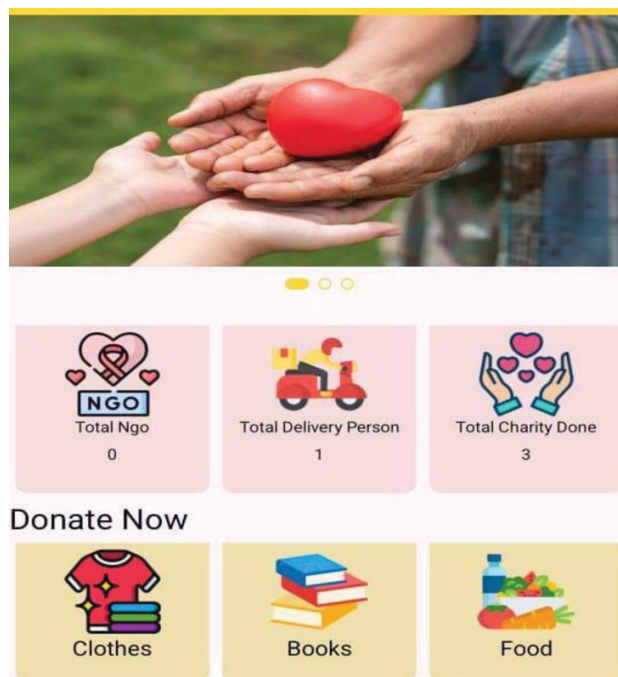


Figure 15.4. Food donation and management dashboard.

Source: Author.

Register As	
Customer	NGO
Delivery Person Name	
Delivery Person Email	
Delivery Person Number	
Delivery Person Address	
Delivery Person Pincode	
☐ Is License Available?	
License Number	
Vehicle Number	
Delivery Password	
REGISTER DELIVERY PERSON	

Figure 15.3. Delivery personnel registration interface.

Source: Author.



Figure 15.5. Food donation record.

Source: Author.

settings, update profile details, reset passwords, and contact support for assistance as displayed in Figure 15.6.

The donation process helps users to easily navigate through the donation form and fill in details. As seen in Figure 15.7, users can fill food name, type, condition, also the preparation time to ensure quality food is being delivered. This allows the delivery person to neatly check and verify whether the food which is said is properly present or not. The system also reminds users to remove the pin after donation to prevent confusion. This feature ensures a smooth and efficient food-sharing process.

A major feature of the system is the food map, which allows users to view available food donation locations. As shown in Figure 15.8, the map shows the location of the user and helps the delivery person to identify the nearby NGO's or the receiver's place.

A delivery person can navigate to these places and update the location to both the user and receiver about the delivery of food or clothes, or books. This feature develops transparency between the Donor and Receiver.

The Android application provides the error feature if a person tries to donate or receive multiple food donations at the same time. As seen in Figure 15.9, if a user attempts to place a PIN when one already exists, the system sends

Figure 15.7. Food donation process.

Source: Author.

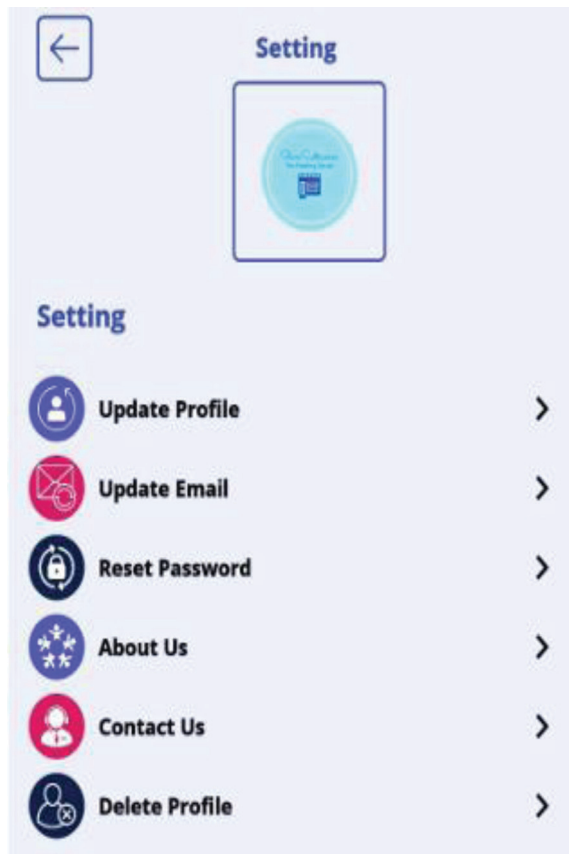


Figure 15.6. Profile management panel.

Source: Author.



Figure 15.8. Location-based food distribution.

Source: Author.

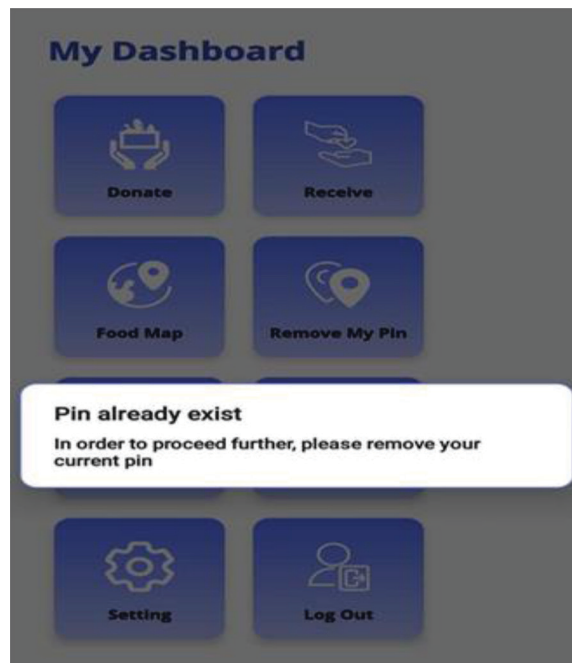


Figure 15.9. Location alert notification.

Source: Author.

an alert message that the PIN already exists in the System so that the user can first remove that PIN and then try to place a new PIN.

6. Conclusion

The food-sharing and tracking system is working perfectly with the frontend and can be used by donors and receivers through a web-based platform if launched. The advantage of this model is that it provides a lightweight and efficient solution for food donation without requiring complex infrastructure. It takes the good points of existing food-sharing methodologies and customizes them as per the need, making it a new type of model on its own, designed specifically for efficient food distribution. This is just the beginning of technology-driven food redistribution, and if developed further with AI-based demand prediction, IoT-enabled food quality monitoring, and blockchain for transparency, it can provide a more robust and scalable solution. This research will leave a mark of its own by

contributing to bridging the gap between food security and technology.

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